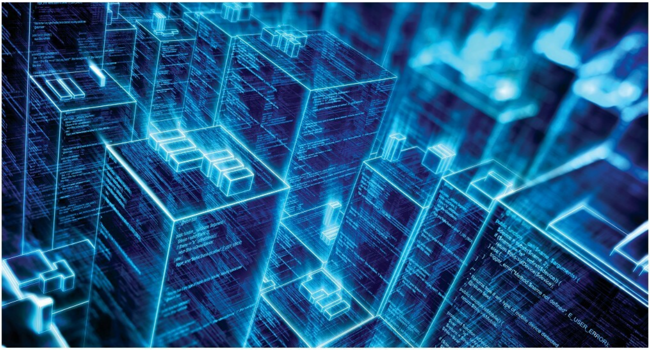
A Hands-on Guide on Virtualisation with VirtualBox

Virtualisation is the process of creating software based (or virtual) representation of a resource rather than a physical one. Virtualisation is applicable at the compute, storage or network levels. In this article we will discuss compute level virtualisation, which is commonly referred to as server virtualisation.



Server virtualisation (henceforth referred to as virtualisation) allows us to run multiple instances of operating systems (OS) simultaneously on a single server. These OSs can be of the same or of different types. For instance, you can run Windows as well as Linux OS on the same server simultaneously. Virtualisation adds a software layer on top of the hardware, which allows users to share physical hardware (memory, CPU, network, storage and so on) with multiple OSs. This virtualisation layer is called the virtual machine manager (VMM) or a hypervisor. There are two types of hypervisors.

Bare metal hypervisors: These are also known as Type-1 hypervisors and are directly installed on hardware. This enables the sharing of hardware resources with a guest OS (henceforth referred to as 'guest') running on top of them. Each guest runs in an isolated environment without interfering with other guests. ESXi, Xen, Hyper-V and KVM are examples of bare metal hypervisors.

Hosted hypervisors: These are also known as Type-2 hypervisors. They cannot be installed directly on hardware. They run as applications and hence require an OS to run them. Similar to bare metal hypervisors, they are able to share physical resources among multiple guests and the physical host on which they are running. VMware Workstation and Oracle VM VirtualBox (hereafter referred to as VirtualBox) are examples of hosted hypervisors.

An introduction to VirtualBox

VirtualBox is cross-platform virtualisation software. It is available on a wide range of platforms like Windows, Linux, Solaris, and so on. It extends the functionality of the existing OS and allows us to run multiple guests simultaneously along with the host's other applications.

VirtualBox terminology

To get a better understanding of VirtualBox, let's get familiar with its terminology.

1) *Host OS:* This is a physical or virtual machine on which VirtualBox is installed.
2) *Virtual machine:* This is the virtual environment created to run the guest OS. All its resources, like the CPU, memory, storage, network devices, etc, are virtual.
3) *Guest OS:* This is the OS running inside VirtualBox. VirtualBox supports a wide range of guests like Windows, Solaris, Linux, Apple, and so on.
4) *Guest additions:* These are additional software bundles